# Page Coloring

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TLB = Translation Lookaside Buffer

# What is a page color

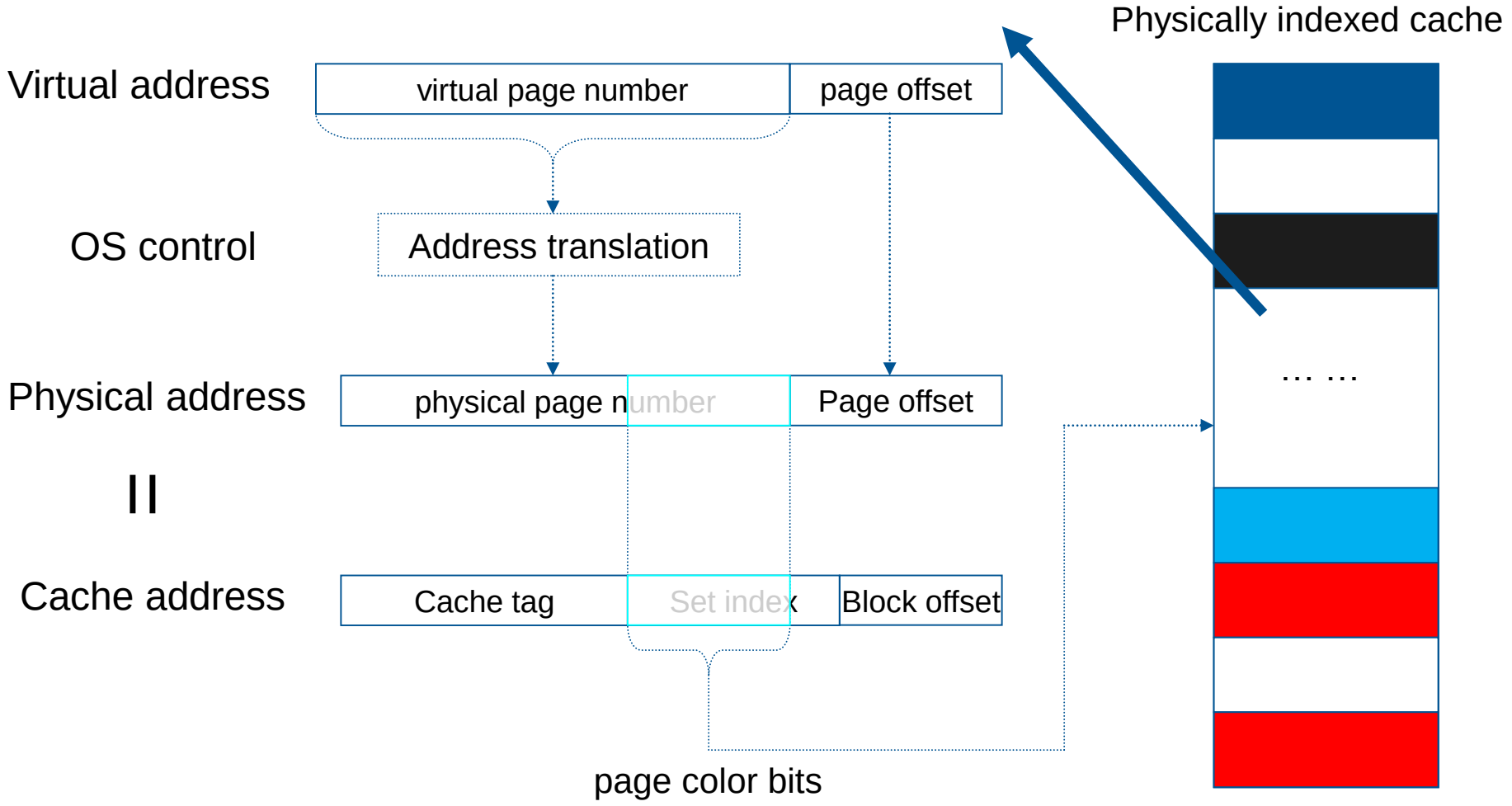
- A cache has multiple sets.
- A certain number of sets put-together constitute page color (also just called “color”).
- Example 1a: A cache has 1024 sets and a color has 32 sets. How many colors are there in this cache?
- Answer: 32.
- Example 1b: In above cache, there are 8 ways and every block is 64B. What is the total storage in a single set?
- Answer: 8*64B = 512B
- Example 1c: What is the total storage in a single color?
- Answer: 32*512B = 16KB

# How to find number of page colors?

- Page color bits = common bits between physical page number and set index

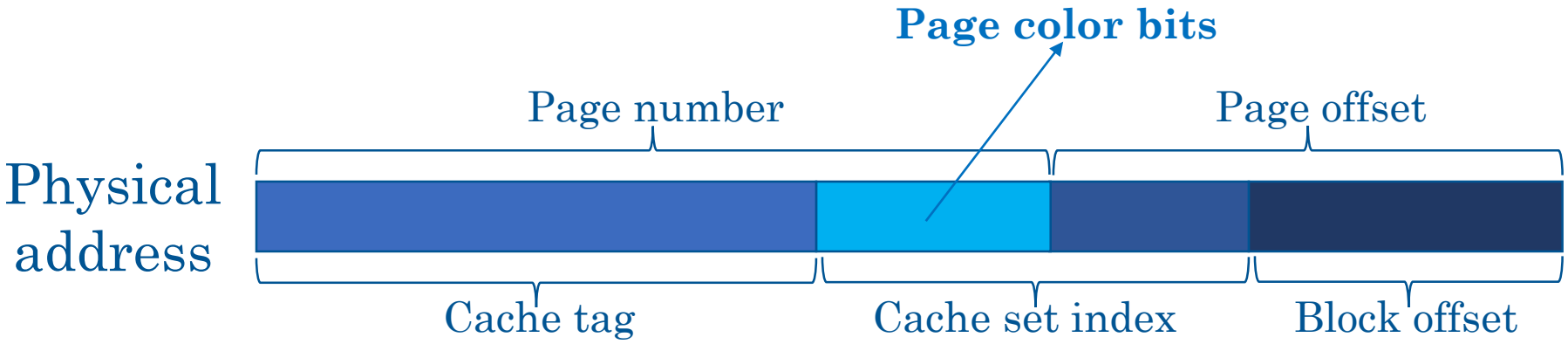
# Page Coloring

- Physically indexed caches are divided into multiple regions (colors).
- All cache lines in a physical page are cached in one of those regions (colors).




Lin et al. HPCA'08

# Computing # of Page colors



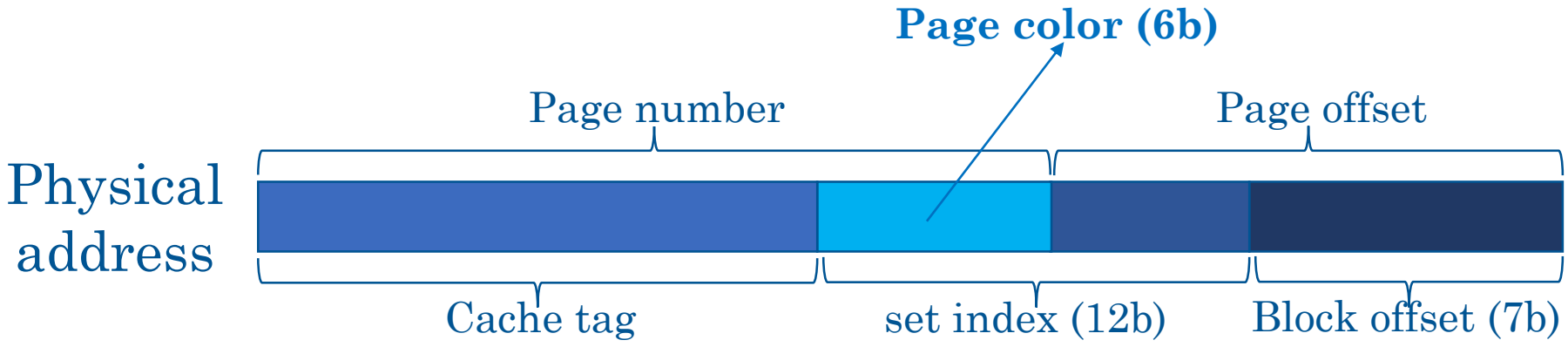

#PageColor bits = # BlockOffset bits + #SetIndex bits - #PageOffset bits

=> #PageColors = (CacheBlockSize * NumberOfSets)/PageSize

= CacheSize/(PageSize * CacheAssociativity)

# Solved example

1. For a cache with 128B block-size, 4096 sets and 64 colors, find the system page size.




2. For a cache with 128B block-size, 4096 sets and page size of 1MB, find number of page colors

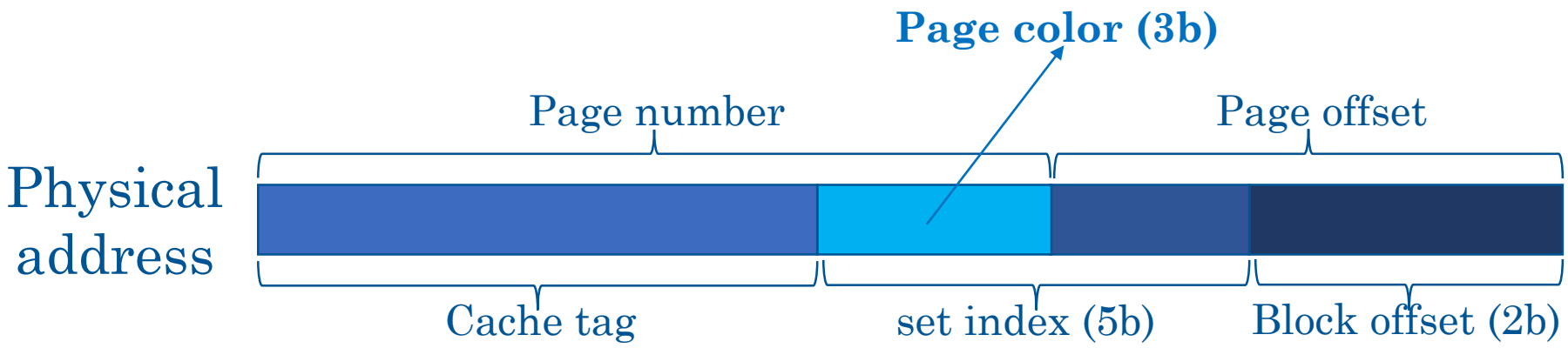
Answer: no page colors

# Example of page-color

- Consider a cache with 4B block size. It has 32 sets and 4-way associativity. Find total cache size.
- Answer: 512B (=32*4*4B)
- If system page-size is 16B, how many page colors are there in the cache.
- Answer: 8 colors are there. There are 3 page-color bits.
- How many sets are there in each color.
- Answer: We have 32 sets and 8 colors. So 4 sets per color.
- Which sets belong to color-number 0
- Answer: 0 to 3

# Example of page-color (continued)

- For every value of page color, “2 lower bits of set-index” can take 4 values. Hence, every page color will have 4 sets.



# Example of page-color (continued)

| Colors | Sets |
|---|---|
| 0 | 0 to 3 |
| 1 | 4 to 7 |
| 2 | 8 to 11 |
| 3 | 12 to 15 |
| 4 | 16 to 19 |
| 5 | 20 to 23 |
| 6 | 24 to 27 |
| 7 | 28 to 31 |

If a set is assigned to a color, then all the ways of this set are assigned to that color. True or false?
True

# Solved Example

- A processor with 32 bit physical address has only L1 cache. L1 cache has 32B block size, has 4 ways and a total size of 16KB.
- The processor uses a page size of 1KB. How many page colors are there in the system?
- Number of sets * block size = 16KB/4 = 4KB
- The page colors = 4KB/1KB = 4

# How OS uses page coloring to limit the amount of memory used by any process

# Example on How Page Coloring works

- Assume a VA 1010101010101010 (16bits). Page size =16B. Cache block size =4B. Number of sets = 64.
- Find page color for this VA and which set will this address be mapped to (if i is a virtual-indexed cache).
- Solution: Page color = 1010, set index = 101010
- Case 1: A TLB has an entry shown below. Find set index which this address be mapped to in a physically-indexed cache.

| VPN | PFN |
|---|---|
| 101010101010 | 101010101111 |

- **Answer**: 111110
- Explanation: For the VA 1010101010101010, PA is 1010101111111010. Removing the last 2 bits for block-offset, next 6 bits give the set-index.

# How Page Coloring works

- Case 2: if a TLB has below entry, find set index which this address be mapped to if it is a physically-indexed cache.

| VPN | PFN |
|---|---|
| 101010101010 | 101010100001 |

- **Answer:** 000110
- Explanation: For the VA 101010101010101010, PA is 101010000011010. Removing the last 2 bits for block-offset, next 6 bits give the set-index
- Lesson: By changing the VPN to PFN entry, the OS can change the mapping of a virtual address to physical address.
- Example: OS maps all VPNs of a process to just few PFNs ➔ this process now has only one page worth of space. So, its memory allocation has been restricted.

# Page Coloring summary

- VA converted to PA by OS-controlled address translation
- PA used in a physically indexed cache.
- Physically indexed cache is divided into multiple regions (colors).
- Each physical page will cached in one of those regions indexed by the page color
- OS can control the page color of a virtual page through address mapping (by selecting a physical page with a specific value in its page color bits).